Algebra II with Trigonometry Chapter 11.2

Olympiad Problems (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file: 4666322_alg-2trig-h-chp-11-2-note-docx.pdf

Problems

1. An ellipse has the equation $\frac{x^2}{169} + \frac{y^2}{144} = 1$. Let F_1 and F_2 be its foci. If P is a point on the ellipse that does not lie on the x-axis, what is the perimeter of $\triangle PF_1F_2$?

2. For a certain ellipse, the length of its minor axis, the distance between its foci, and the length of its major axis form an arithmetic progression in strictly increasing order. What is the eccentricity of this ellipse?

3. A point P lies on the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$. If the distance from P to the origin is exactly 4, what is the product of the distances from P to the two foci of the ellipse?

4. The ellipses $\frac{x^2}{20} + \frac{y^2}{5} = 1$ and $\frac{x^2}{5} + \frac{y^2}{20} = 1$ intersect at four distinct points. What is the area of the convex quadrilateral whose vertices are these four intersection points?

5. For a certain ellipse centered at the origin, a circle centered at the origin passes through both of the ellipse's foci as well as both endpoints of its minor axis. Evaluate the eccentricity of the ellipse.

6. An ellipse centered at the origin with its axes parallel to the coordinate axes passes through the points $(6, 4)$ and $(8, 3)$. What is the eccentricity of this ellipse?